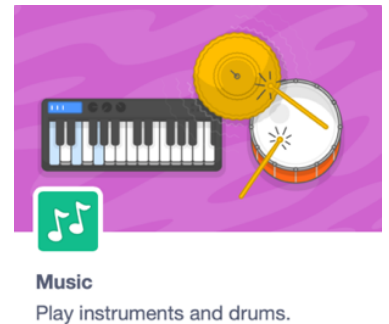


Scratch Looper



Register/login at <https://scratch.mit.edu>

Make rhythms by looping.

1) **Create** a new project and add the Music extension.

*A **Beat** is a steady, recurring pulse in music that you can tap your foot to.*

***Tempo** is the speed at which sounds play, measured in BPM (beats per minute). The starting tempo in Scratch is **60 BPM**, which means 1 beat each second.*

1) Create code that **plays drums** for one beat, or 1 second. Start the code with the **green flag** and use a **forever** loop to make it repeat.

*Do you hear one beat per second?
This is your “rhythm section.”*

*A **Bar** splits music into small, equal groups of beats. A lot of pop music has **4 beats** in every bar.*



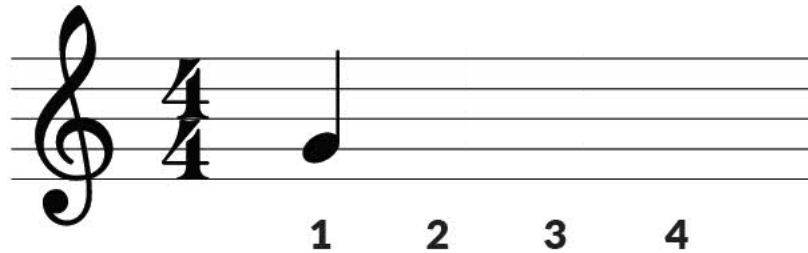
2) Add a new code block that also starts on the **green flag**. Make this code **repeat 4** beats to make a bar.

3) **Set an instrument** before you enter the loop and **play a note** for 1 beat, staying in time with the drums.

Musicians call the first beat of the bar “**The One.**”

- 4) To play on “The One” you’ll need to count the beats in the bar. Create a loop variable called **beats**.
- 5) To count 1, 2, 3, 4, set **beats** to 1 before you enter the loop and **change beat by 1** as the last thing in the loop.

- 6) Add an **if .. else** around the **note** so that it only plays on “The One.”



Otherwise (**else**) add a **rest** for 1 beat.

A **rest** is a silent delay.

- 7) We want the bar to repeat so put this loop and the setup of the loop variable inside another **forever** loop.

This is a **nested loop** as one loop is nested inside another.

- 8) Now, add some different notes on other beats in the bar. Add another **if .. else** replacing the **rest**, and check for a different beat.
- 9) Don’t forget to put the **rest** back in at the last **else** in the chain.

Have fun looping! You can always change the tempo by dragging a **set tempo** block onto your code space and just clicking on it with a different value.

Save your code with a good name.

File > Save now